

Weekly Progress Report: TruthLens AR Bangla Multimodal Fake News and Deepfake Detection

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Abstract—This report summarizes the weekly progress of the TruthLens AR project. The team completed the literature review, initiated dataset collection, benchmarked baseline models, and finalized the proposed multimodal architecture integrating Vision-Language Models with ARCore.

Index Terms—Fake News Detection, Deepfake Detection, Vision-Language Models, Augmented Reality, ARCore, Bangla NLP, Multimodal Learning

I. INTRODUCTION

TruthLens AR is a multimodal misinformation detection system for Bangla content that combines Vision-Language Models (VLMs) with Augmented Reality (AR). Unlike conventional text-only approaches, the proposed framework jointly analyzes textual and visual information and displays credibility results in real time using Google ARCore. This week focused on establishing the research foundation, preparing datasets, benchmarking baseline models, and finalizing the proposed architecture.

II. COMPLETED TASKS

A. Literature Review

The team reviewed recent literature on Bangla fake news detection, multimodal learning, transformer-based language models, Vision-Language Models, and AR-assisted information verification.

Major findings include:

- Existing Bangla fake news detectors rely primarily on textual analysis.
- Few approaches verify semantic consistency between text and accompanying images.
- Real-time AR-based misinformation visualization remains largely unexplored.
- Current regional datasets contain limited multimodal image-text pairs.

These observations clearly establish the research gap addressed by TruthLens AR.

B. Dataset Collection

Dataset preparation activities were initiated by selecting the BanFakeNews corpus as the primary textual dataset. The team began cleaning the dataset, removing duplicate samples, validating annotations, and organizing authentic and fake news into structured directories. Corresponding image collection and image-text alignment have also begun to support multimodal learning.

C. Existing Model Benchmarking

Several baseline transformer models were successfully downloaded and executed within the local development environment. Software dependencies were installed, GPU compatibility was verified, and preliminary inference tests were completed successfully. These baseline models will serve as performance references during future evaluation of the TruthLens framework.

D. Proposed Solution

The overall architecture of TruthLens AR has been finalized. The proposed workflow consists of the following steps:

- 1) Capture live video frames.
- 2) Extract Bangla text using OCR.
- 3) Process images and text using cloud-hosted Vision-Language Models.
- 4) Perform multimodal semantic verification.
- 5) Return credibility prediction.
- 6) Display color-coded verification overlays using Google ARCore.

The finalized technology stack includes Android Studio, Kotlin, Python, PyTorch, Hugging Face Transformers, Easy-OCR, Google ARCore, FastAPI, and cloud-hosted VLMs.

III. CHALLENGES

The primary challenge involves minimizing cloud-to-mobile inference latency while maintaining high prediction accuracy. Additional challenges include limited Bangla multimodal datasets, OCR inaccuracies on noisy images, smartphone resource constraints, and maintaining stable AR overlays under varying environmental conditions.

IV. NEXT WEEK'S PLAN

The team will begin implementation of the integrated prototype by connecting OCR, Vision-Language Models, backend APIs, and ARCore anchors. Planned activities include fine-tuning the VLM using Bangla datasets, optimizing inference latency, integrating cloud services, and conducting end-to-end testing of the complete verification pipeline.

V. CONCLUSION

During this reporting period, the team completed the literature review, initiated dataset preparation, benchmarked baseline models, and finalized the TruthLens AR architecture. These milestones establish a strong foundation for system integration and prototype development in the next phase.